

Professional Computing Ethics (PCE) — Mock Exam Model Answers

Grounded in Kenneth C. Laudon & Jane P. Laudon, Management Information Systems, 13th ed. (Chapter 4, "Ethical and Social Issues in Information Systems," and Chapter 8, "Securing Information Systems"). Note: this edition predates GDPR (2018) and the recent AI-fairness literature, so questions on GDPR, algorithmic bias, AI fairness, and explainability are answered using standard course material while connecting back to the textbook's principles (Fair Information Practices, privacy, and the candidate ethical principles).

1. Discuss the importance of IT ethics in today's digital age, with real-life examples, and explain how ethical behaviour benefits individuals and organisations.

Ethics refers to the principles of right and wrong that individuals, acting as free moral agents, use to make choices to guide their behaviour. In the digital age, information technology has *heightened* ethical concerns because, as Laudon notes, technology trends such as exponentially growing computing power, cheap data storage, advanced data analysis, and pervasive networking make it cheap and effective to collect, combine, and exploit detailed information about people (a practice the textbook calls **profiling**). IT ethics matters because these systems create new dilemmas where one set of interests is pitted against another, and existing laws often lag behind the technology.

Real-life examples:

- **Facebook-Cambridge Analytica:** personal data was harvested and used for political targeting without genuine informed consent — a violation of privacy and consent principles.
- **Employee monitoring:** companies track employees' email and internet use to protect productivity, raising a conflict between the firm's interests and employees' privacy (an example Laudon uses directly).
- **Data breaches** (e.g., Equifax) where poor security exposed millions of records, harming individuals through identity theft.

Benefits of ethical behaviour:

- *For individuals:* protects privacy, dignity, and safety; builds trust; avoids legal and reputational harm.
- *For organisations:* builds customer trust and loyalty, avoids lawsuits and regulatory fines, protects brand reputation, and improves decision quality. As Laudon warns, "the appearance of unethical behavior may do as much harm to you and your company as actual unethical behavior."

2. Explain the 4 key ethical principles of data science, with an example for each.

The widely taught framework for data-science ethics is often summarised as **Fairness, Accountability, Transparency, and Privacy** (sometimes "FAT" + Privacy). These map onto the textbook's moral dimensions of accountability/control and information rights.

1. **Fairness** — data and models should not discriminate against or disadvantage any group.
Example: a credit-scoring model must not deny loans based on race or gender, even indirectly through proxy variables.
2. **Accountability** — those who build and deploy systems must take responsibility for their outcomes and be answerable when harm occurs. This reflects Laudon's moral dimension of **accountability and control** ("Who can and will be held accountable for harm done?").
Example: if a self-driving car's algorithm causes a crash, the company must have clear responsibility and a remediation process.
3. **Transparency** — how data is collected, processed, and used should be open and understandable, not hidden. *Example:* a company publishing a clear privacy policy and disclosing that it uses an algorithm to rank job applicants.
4. **Privacy / data protection** — personal data must be collected and used lawfully, with consent, and kept secure. This directly reflects Laudon's **Fair Information Practices**.
Example: a health app encrypting patient records and only collecting the minimum data needed.

3. How does an ethical framework or guideline differ from a regulation? Explain with examples.

An **ethical framework/guideline** is a set of *voluntary* principles that help people decide what is right; it is self-imposed and enforced by professional norms or conscience, not by law. A **regulation** is a *legally binding* rule made by a government or authority, enforced through penalties such as fines or imprisonment.

In Laudon's terms, **professional codes of conduct** (e.g., the **ACM Code of Ethics**, with imperatives such as avoiding harm, honouring property rights, and respecting privacy) are guidelines — "promises by professions to regulate themselves in the general interest of society." By contrast, laws like **HIPAA** (medical-record privacy) or the **Children's Online Privacy Protection Act (COPPA)** are regulations carrying legal force.

Examples:

- *Guideline:* The ACM Code of Ethics asks computing professionals to "avoid harm" — but breaking it brings professional, not legal, consequences.
- *Regulation:* The EU's data protection law (the Directive, later **GDPR**) legally requires consent before processing personal data, and violations carry heavy fines.

In short: guidelines tell you what you *should* do; regulations tell you what you *must* do.

4. Explain algorithmic bias, with 2 examples.

Algorithmic bias occurs when a computer system produces systematically unfair or discriminatory outcomes, typically because it was trained on biased or unrepresentative data, or because its design encodes human prejudices. The algorithm appears "objective" but actually reproduces and can amplify existing inequalities. This connects to Laudon's concern with **system quality** (what standards of data and system quality protect individual rights?).

Examples:

1. **Amazon's recruiting tool:** an AI built to screen résumés was trained on a decade of mostly male applications, so it learned to penalise résumés containing the word "women's" (e.g., "women's chess club"), discriminating against female candidates. Amazon scrapped it.
2. **COMPAS recidivism algorithm:** a tool used in U.S. courts to predict whether a defendant would re-offend was found to falsely flag Black defendants as high-risk at roughly twice the rate of white defendants, leading to unfair sentencing recommendations.

(Other commonly cited cases: facial-recognition systems that perform worse on darker-skinned faces and women.)

5. Why is fairness an important principle in AI ethics?

Fairness matters because AI systems increasingly make or influence high-stakes decisions — hiring, lending, policing, healthcare — that directly affect people's lives and opportunities. If a system is unfair, it can:

- **Cause real harm and discrimination** against individuals or whole groups (as in the COMPAS and Amazon examples above).
- **Amplify existing inequalities at scale** — unlike a single biased human, a biased algorithm applies the same unfair rule to millions of cases automatically.
- **Erode trust** in technology and in the organisations using it.
- **Create legal and reputational liability** — discrimination may breach anti-discrimination law and data-protection regulation.

Fairness also underpins the legitimacy of AI: people will only accept automated decisions if they believe those decisions treat everyone equitably. It aligns with the textbook's **Golden Rule** ("do unto others as you would have them do unto you") and the moral dimension of **quality of life** — preserving fair treatment as a core social value in an information society.

6. What is GDPR? Choose and explain 3 user rights stated in GDPR.

The **General Data Protection Regulation (GDPR)** is a European Union regulation (in force since May 2018) that governs how organisations collect, store, and process the personal data of EU

residents. It is the successor to the **EU Directive on Data Protection** that Laudon describes, which already required companies to inform people when collecting data, obtain **informed consent**, and allow access and correction. GDPR strengthened these rules, applies to any organisation handling EU residents' data (even outside the EU), and imposes large fines for violations.

Three key user (data-subject) rights under GDPR:

1. **Right of access:** Individuals can ask an organisation what personal data it holds about them and how it is being used, and obtain a copy. (*This extends Laudon's "access/participation" FIP principle.*)
2. **Right to erasure ("right to be forgotten"):** Individuals can request that their personal data be deleted when it is no longer needed or consent is withdrawn.
3. **Right to rectification:** Individuals can require that inaccurate or incomplete personal data about them be corrected.

(Other GDPR rights include data portability, the right to object to processing, and the right to restrict processing.)

7. In data collection, what is informed consent? What are its 4 elements? Provide 2 examples.

Laudon defines **informed consent** as "consent given with knowledge of all the facts needed to make a rational decision." In data collection, it means a person must genuinely understand and freely agree to how their personal data will be collected and used *before* it is collected.

Four elements of informed consent:

1. **Disclosure** — the person is told all relevant facts (what data is collected, why, how it will be used, and who will receive it). This mirrors the FIP **notice/awareness** principle.
2. **Comprehension / understanding** — the information is presented clearly so the person actually understands it (not buried in legal jargon).
3. **Voluntariness** — consent is given freely, without coercion or pressure.
4. **Competence / capacity** — the person is capable of making the decision (e.g., an adult of sound mind; this is why COPPA requires parental consent for children under 13).

Examples:

1. A website shows a clear cookie banner explaining what data it tracks and lets the user **opt in** before any tracking begins (the stricter "opt-in" model Laudon contrasts with "opt-out").
2. A medical research study gives participants a plain-language consent form explaining how their health data will be used, and they sign it voluntarily after asking questions.

8. What is the CIA Triad in data protection? Briefly explain each of the 3 elements.

The **CIA Triad** is the foundational model of information security, made up of three goals every secure system must uphold. (Laudon's definition of **security** — "the policies, procedures, and technical measures used to prevent unauthorized access, alteration, theft, or physical damage to information systems" — maps directly onto these three goals.)

1. **Confidentiality** — ensuring that data is accessible only to authorised people and protected from unauthorised disclosure. *Mechanisms*: encryption, access controls, authentication. *Threat*: a data breach or sniffer program stealing private records.
2. **Integrity** — ensuring data is accurate, complete, and not altered by unauthorised parties. *Mechanisms*: checksums, hashing, version control. *Threat*: a hacker tampering with bank balances or a virus corrupting files.
3. **Availability** — ensuring that systems and data are accessible to authorised users when needed. *Mechanisms*: backups, redundancy, failover. *Threat*: a **denial-of-service (DoS) attack** that floods a server so it cannot serve legitimate users (a threat Laudon describes in detail).

9. What are the various threats to data security? Explain at least 3.

Laudon's Chapter 8 catalogues many threats. At least three:

1. **Malicious software (malware)**: Programs designed to harm systems, including:
 - *Viruses* — rogue code that attaches to files and executes without permission, often carrying a destructive "payload," spreading when users share infected files.
 - *Worms* — independent programs that copy themselves across networks without human action, spreading faster than viruses.
 - *Trojan horses* — programs that appear benign but do something harmful, often a gateway for other malware.
2. **Phishing and spoofing**: **Spoofing** is misrepresenting identity (e.g., fake email addresses or redirecting users to a fraudulent website). **Phishing** sets up fake websites or sends fake emails imitating legitimate businesses to trick users into giving up confidential data such as passwords or card numbers. Variants include *spear phishing*, *evil twins* (fake Wi-Fi hotspots), and *pharming*.
3. **Denial-of-Service (DoS) attacks**: Hackers flood a server with thousands of false requests so it crashes and cannot serve legitimate users. A **distributed DoS (DDoS)** uses many computers at once to overwhelm the target.

(Other threats Laudon discusses: *sniffers/eavesdropping*, *SQL injection attacks*, *identity theft*, and *click fraud*.)

10. Why is data privacy important in software development, and how should a developer handle it? Provide 3 best-practice examples.

Why it matters: Software increasingly collects and processes personal data, and Laudon stresses that advances in storage and analysis have made the "routine violation of individual privacy both cheap and effective." Poor privacy handling harms users (identity theft, surveillance, loss of dignity), exposes the organisation to legal penalties and reputational damage, and breaches Fair Information Practices and regulations like GDPR/HIPAA.

How a developer should handle it: Treat privacy as a design requirement, not an afterthought — the FTC's "**privacy by design**" principle (which Laudon cites). Collect only what is necessary, secure it, be transparent, and obtain consent.

Three best practices for data protection:

1. **Data minimisation and purpose limitation:** Collect only the data the application genuinely needs, and use it only for the stated purpose. *Example:* a weather app should not request access to the user's contacts.
2. **Encryption and secure storage/transmission:** Protect data at rest and in transit so that even if intercepted it cannot be read. *Example:* using HTTPS/TLS and encrypting stored passwords (hashing + salting) rather than keeping them in plain text.
3. **Access control and authentication:** Restrict data access to authorised users only, using strong authentication and the principle of least privilege. *Example:* role-based access so only authorised staff can view customer records, plus logging of who accessed what.

(Additional good practices: transparent privacy notices, obtaining informed/opt-in consent, allowing users to access and delete their data, and regular security testing.)

11. What is the explainability of an AI system?

Explainability (also called interpretability, or "Explainable AI / XAI") is the degree to which a human can understand *why* an AI system produced a particular decision or output. A highly explainable system can show the reasons or factors behind its result, whereas a "black box" system (such as a deep neural network) gives an answer with no clear justification.

Explainability is important because:

- It supports **accountability** — if we can see why a decision was made, we can hold someone responsible for errors (Laudon's accountability-and-control dimension).
- It enables **fairness and bias detection** — we can check whether a decision relied on inappropriate factors.
- It builds **trust** — users and regulators are more willing to accept automated decisions they can understand.

- It is increasingly a **legal requirement** — for example, GDPR gives individuals a right to meaningful information about automated decision-making that affects them.

Example: a loan-rejection system that tells the applicant "declined because of insufficient income and a short credit history" is explainable; one that simply says "declined" is not.

12. How do you plan to implement PCE concepts in your future work? Provide 3 examples.

Applying Professional Computing Ethics in practice means embedding ethical thinking into everyday technical decisions. Three concrete ways:

1. **Build privacy and security in by design.** When developing any application, I will follow Fair Information Practices and "privacy by design": collect only necessary data, obtain clear informed consent, encrypt sensitive information, and apply strong access controls — protecting confidentiality, integrity, and availability (the CIA triad). *Example:* before adding a feature that collects user location, I would ask whether it is truly needed, disclose it clearly, and let users opt in.
2. **Test for fairness and bias in data and algorithms.** Before deploying any model that affects people, I will check the training data for representativeness and test outputs for discriminatory patterns, and I will favour explainable approaches so decisions can be justified. *Example:* auditing a hiring or scoring model for biased outcomes across gender or ethnicity before release, as the Amazon recruiting-tool case warns.
3. **Apply ethical decision-making and professional codes when facing dilemmas.** When competing interests arise, I will use Laudon's candidate ethical principles — the Golden Rule, the Categorical Imperative ("if everyone did this, could society survive?"), the slippery-slope rule, the Utilitarian and Risk-Aversion principles — and follow professional codes such as the ACM Code of Ethics. *Example:* if asked to repurpose customer data for marketing without consent, I would recognise it as an FIP/consent violation, raise the concern, and propose an opt-in alternative rather than proceed.

In short, I will treat ethics not as a separate step but as part of accountable, trustworthy professional practice throughout the software lifecycle.